

First, we would take a cone and a hemisphere and bring their flat faces together. Here, of course, we would take the base radius of the cone equal to the radius of the hemisphere, for the toy is to have a smooth surface. So, the steps would be as shown in Fig. 12.5.

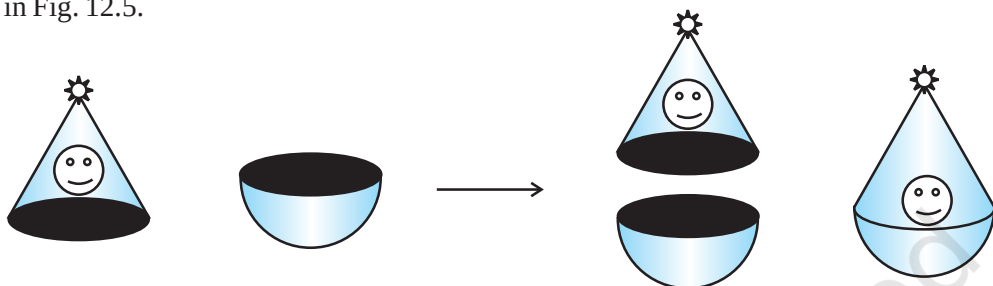


Fig. 12.5

At the end of our trial, we have got ourselves a nice round-bottomed toy. Now if we want to find how much paint we would require to colour the surface of this toy, what would we need to know? We would need to know the surface area of the toy, which consists of the CSA of the hemisphere and the CSA of the cone.

So, we can say:

$$\text{Total surface area of the toy} = \text{CSA of hemisphere} + \text{CSA of cone}$$

Now, let us consider some examples.

Example 1 : Rasheed got a playing top (*lattu*) as his birthday present, which surprisingly had no colour on it. He wanted to colour it with his crayons. The top is shaped like a cone surmounted by a hemisphere (see Fig 12.6). The entire top is 5 cm in height and the diameter of the top is 3.5 cm. Find the area he

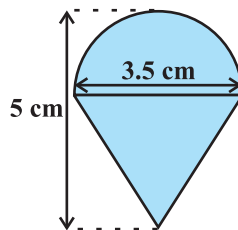


Fig. 12.6

has to colour. (Take $\pi = \frac{22}{7}$)

Solution : This top is exactly like the object we have discussed in Fig. 12.5. So, we can conveniently use the result we have arrived at there. That is :

$$\text{TSA of the toy} = \text{CSA of hemisphere} + \text{CSA of cone}$$

$$\text{Now, the curved surface area of the hemisphere} = \frac{1}{2}(4\pi r^2) = 2\pi r^2$$

$$= \left(2 \times \frac{22}{7} \times \frac{3.5}{2} \times \frac{3.5}{2} \right) \text{cm}^2$$

Also, the height of the cone = height of the top – height (radius) of the hemispherical part

$$= \left(5 - \frac{3.5}{2} \right) \text{ cm} = 3.25 \text{ cm}$$

So, the slant height of the cone (l) = $\sqrt{r^2 + h^2} = \sqrt{\left(\frac{3.5}{2}\right)^2 + (3.25)^2} \text{ cm} = 3.7 \text{ cm (approx.)}$

$$\text{Therefore, CSA of cone} = \pi r l = \left(\frac{22}{7} \times \frac{3.5}{2} \times 3.7 \right) \text{ cm}^2$$

This gives the surface area of the top as

$$\begin{aligned} &= \left(2 \times \frac{22}{7} \times \frac{3.5}{2} \times \frac{3.5}{2} \right) \text{ cm}^2 + \left(\frac{22}{7} \times \frac{3.5}{2} \times 3.7 \right) \text{ cm}^2 \\ &= \frac{22}{7} \times \frac{3.5}{2} (3.5 + 3.7) \text{ cm}^2 = \frac{11}{2} \times (3.5 + 3.7) \text{ cm}^2 = 39.6 \text{ cm}^2 \text{ (approx.)} \end{aligned}$$

You may note that ‘total surface area of the top’ is *not* the sum of the total surface areas of the cone and hemisphere.

Example 2 : The decorative block shown in Fig. 12.7 is made of two solids — a cube and a hemisphere. The base of the block is a cube with edge 5 cm, and the hemisphere fixed on the top has a diameter of 4.2 cm. Find the total surface area of the block. (Take $\pi = \frac{22}{7}$)

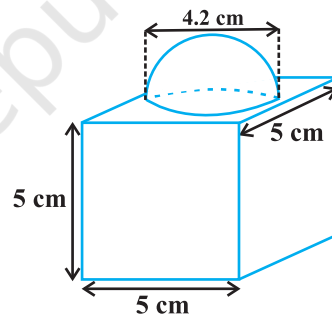


Fig. 12.7

Solution : The total surface area of the cube = $6 \times (\text{edge})^2 = 6 \times 5 \times 5 \text{ cm}^2 = 150 \text{ cm}^2$.

Note that the part of the cube where the hemisphere is attached is not included in the surface area.

So, the surface area of the block = TSA of cube – base area of hemisphere + CSA of hemisphere

$$= 150 - \pi r^2 + 2 \pi r^2 = (150 + \pi r^2) \text{ cm}^2$$

$$= 150 \text{ cm}^2 + \left(\frac{22}{7} \times \frac{4.2}{2} \times \frac{4.2}{2} \right) \text{ cm}^2$$

$$= (150 + 13.86) \text{ cm}^2 = 163.86 \text{ cm}^2$$

Example 3 : A wooden toy rocket is in the shape of a cone mounted on a cylinder, as shown in Fig. 12.8. The height of the entire rocket is 26 cm, while the height of the conical part is 6 cm. The base of the conical portion has a diameter of 5 cm, while the base diameter of the cylindrical portion is 3 cm. If the conical portion is to be painted orange and the cylindrical portion yellow, find the area of the rocket painted with each of these colours. (Take $\pi = 3.14$)

Solution : Denote radius of cone by r , slant height of cone by l , height of cone by h , radius of cylinder by r' and height of cylinder by h' . Then $r = 2.5$ cm, $h = 6$ cm, $r' = 1.5$ cm, $h' = 26 - 6 = 20$ cm and

$$l = \sqrt{r^2 + h^2} = \sqrt{2.5^2 + 6^2} \text{ cm} = 6.5 \text{ cm}$$

Here, the conical portion has its circular base resting on the base of the cylinder, but the base of the cone is larger than the base of the cylinder. So, a part of the base of the cone (a ring) is to be painted.

So, the area to be painted orange = CSA of the cone + base area of the cone
– base area of the cylinder

$$\begin{aligned} &= \pi r l + \pi r^2 - \pi (r')^2 \\ &= \pi [(2.5 \times 6.5) + (2.5)^2 - (1.5)^2] \text{ cm}^2 \\ &= \pi [20.25] \text{ cm}^2 = 3.14 \times 20.25 \text{ cm}^2 \\ &= 63.585 \text{ cm}^2 \end{aligned}$$

Now, the area to be painted yellow = CSA of the cylinder
+ area of one base of the cylinder

$$\begin{aligned} &= 2\pi r' h' + \pi (r')^2 \\ &= \pi r' (2h' + r') \\ &= (3.14 \times 1.5) (2 \times 20 + 1.5) \text{ cm}^2 \\ &= 4.71 \times 41.5 \text{ cm}^2 \\ &= 195.465 \text{ cm}^2 \end{aligned}$$

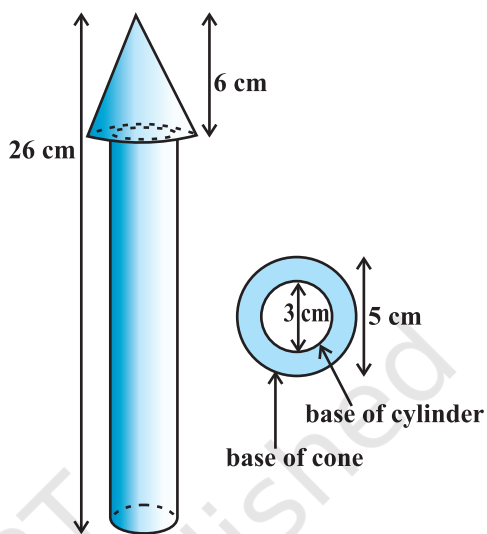


Fig. 12.8

7. A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1 m and 4 m respectively, and the slant height of the top is 2.8 m, find the area of the canvas used for making the tent. Also, find the cost of the canvas of the tent at the rate of ₹500 per m^2 . (Note that the base of the tent will not be covered with canvas.)
8. From a solid cylinder whose height is 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid to the nearest cm^2 .
9. A wooden article was made by scooping out a hemisphere from each end of a solid cylinder, as shown in Fig. 12.11. If the height of the cylinder is 10 cm, and its base is of radius 3.5 cm, find the total surface area of the article.

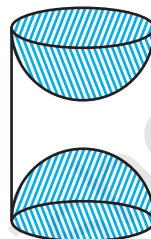


Fig. 12.11

12.3 Volume of a Combination of Solids

In the previous section, we have discussed how to find the surface area of solids made up of a combination of two basic solids. Here, we shall see how to calculate their volumes. It may be noted that in calculating the surface area, we have not added the surface areas of the two constituents, because some part of the surface area disappeared in the process of joining them. However, this will not be the case when we calculate the volume. The volume of the solid formed by joining two basic solids will actually be the sum of the volumes of the constituents, as we see in the examples below.

Example 5 : Shanta runs an industry in a shed which is in the shape of a cuboid surmounted by a half cylinder (see Fig. 12.12). If the base of the shed is of dimension $7\text{ m} \times 15\text{ m}$, and the height of the cuboidal portion is 8 m, find the volume of air that the shed can hold. Further, suppose the machinery in the shed occupies a total space of 300 m^3 , and there are 20 workers, each of whom occupy about 0.08 m^3 space on an average. Then, how much air is in the

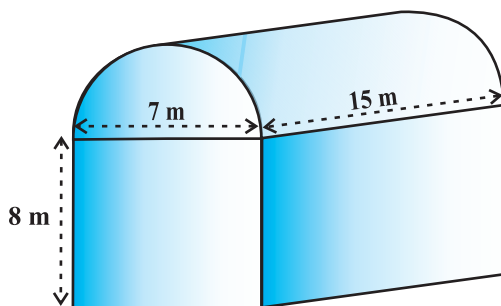


Fig. 12.12

shed? (Take $\pi = \frac{22}{7}$)

Solution : The volume of air inside the shed (when there are no people or machinery) is given by the volume of air inside the cuboid and inside the half cylinder, taken together.

Now, the length, breadth and height of the cuboid are 15 m, 7 m and 8 m, respectively. Also, the diameter of the half cylinder is 7 m and its height is 15 m.

So, the required volume = volume of the cuboid + $\frac{1}{2}$ volume of the cylinder

$$= \left[15 \times 7 \times 8 + \frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 15 \right] \text{ m}^3 = 1128.75 \text{ m}^3$$

Next, the total space occupied by the machinery = 300 m^3

And the total space occupied by the workers = $20 \times 0.08 \text{ m}^3 = 1.6 \text{ m}^3$

Therefore, the volume of the air, when there are machinery and workers

$$= 1128.75 - (300.00 + 1.60) = 827.15 \text{ m}^3$$

Example 6 : A juice seller was serving his customers using glasses as shown in Fig. 12.13. The inner diameter of the cylindrical glass was 5 cm, but the bottom of the glass had a hemispherical raised portion which reduced the capacity of the glass. If the height of a glass was 10 cm, find the apparent capacity of the glass and its actual capacity. (Use $\pi = 3.14$.)

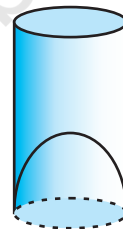


Fig. 12.13

Solution : Since the inner diameter of the glass = 5 cm and height = 10 cm,

the apparent capacity of the glass = $\pi r^2 h$

$$= 3.14 \times 2.5 \times 2.5 \times 10 \text{ cm}^3 = 196.25 \text{ cm}^3$$

But the actual capacity of the glass is less by the volume of the hemisphere at the base of the glass.

$$\text{i.e., it is less by } \frac{2}{3} \pi r^3 = \frac{2}{3} \times 3.14 \times 2.5 \times 2.5 \times 2.5 \text{ cm}^3 = 32.71 \text{ cm}^3$$

So, the actual capacity of the glass = apparent capacity of glass – volume of the hemisphere

$$= (196.25 - 32.71) \text{ cm}^3$$

$$= 163.54 \text{ cm}^3$$